

Instability #21: Honest Boundary

What Is Derived, What Is a Dimensional Error, and What Remains for
the Algebraic Proof

A precise statement of the gap between $|\beta| = -2.197$ (derived) and $d_s/(2 \ln 2) = 2.164$
(conjecture)

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Abstract

CRF AI proposed the algebraic identity $f_{12} \rightarrow d_s/2$ and “bracket $\rightarrow 1/\ln 2$ ” to close Instability #21. The first claim is confirmed and derived. The second contains a dimensional error: the correct statement is bracket $\rightarrow \kappa/\ln 2$, not $1/\ln 2$ (which would give $\beta = -14.4$, not -2.2).

With this correction, the proposed identity is:

$$\beta = \underbrace{\frac{d_s}{2}}_{f_{12}} \cdot \frac{\kappa/\ln 2}{-\kappa} = -\frac{d_s}{2 \ln 2}$$

and the remaining question is whether bracket $= \kappa/\ln 2$ is derivable.

Current status: bracket $= 0.2197$ (measured) vs $\kappa/\ln 2 = 0.2164$ (conjecture) — a 1.5% gap. This is consistent with finite- N noise of order $(n_d/N)^{1/d_s} \approx 0.25$ but is not yet proven to vanish. Instability #21 status: bracket identity is **Watching**, not Closed. The framework has derived all structural components; the final algebraic step is bracket $\rightarrow \kappa/\ln 2$ as $N \rightarrow \infty$.

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1 The Algebraic Identity and the Dimensional Error

CRF AI Proposal (Session, March 2026)

$f_{12} \rightarrow d_s/2 = 1.5$ and $\text{bracket} \rightarrow 1/\ln 2 = 1.4427$
giving $\beta = f_{12} \cdot \text{bracket} = d_s/(2 \ln 2)$.

The formula for β in V20.3 is:

$$\beta = \frac{f_{12} \cdot \text{bracket}}{-\kappa}, \quad \kappa = 0.15$$

Substituting the CRF AI proposal with $\text{bracket} = 1/\ln 2$:

$$\beta = \frac{1.5 \times 1.4427}{-0.15} = -14.4 \neq -2.212$$

This is a factor-of- κ dimensional error. The correct statement is:

Corrected Algebraic Identity

$$\beta = \frac{d_s}{2} \cdot \frac{\kappa/\ln 2}{-\kappa} = -\frac{d_s}{2 \ln 2}$$

requires:

$$f_{12} \rightarrow \frac{d_s}{2} \tag{1}$$

$$\text{bracket} \rightarrow \frac{\kappa}{\ln 2} = \frac{0.15}{\ln 2} = 0.2164 \tag{2}$$

The factor κ is present in the bracket, not in f_{12} . It cancels against the κ in the denominator, yielding $\beta = -d_s/(2 \ln 2)$.

2 What Is Derived

2.1 $f_{12} = d_s/2$: Derived

Derived: $f_{12} = \frac{3}{2} = \frac{d_s}{2}$

From Kuramoto coupling in V20.3 ($K_{\text{eff}} = 0.0069 < K_c = 0.0637$): the system is sub-threshold, phase differences are uniform, $\phi_{\text{var}} = \frac{1}{2}$, $f_{12} = 1 + \frac{1}{2} = \frac{3}{2}$.

In general d_s dimensions: equipartition of phase over d_s orientations gives $\phi_{\text{var}} = d_s/2 - 1$, so

$$f_{12} = 1 + (d_s/2 - 1) = d_s/2$$

For $d_s = 3$: $f_{12} = 3/2$. Measured: 1.527. Error: 1.3%. ✓

2.2 $H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} \| U)$: Derived

From the Born rule and neighbourhood locking (CRF_21_Gaps_ab.pdf, §2): $H_{\text{after}} = 1.799$ (derived) vs 1.816 (measured). Error: 0.9%. ✓

2.3 $\beta = -2.197$: Derived (from derived quantities)

Derived: $\beta = -2.197$ (0.7% from measured -2.212)

$$\beta = \frac{f_{12} \cdot \text{bracket}_{\text{meas}}}{-\kappa} = \frac{\frac{3}{2} \times 0.2197}{-0.15} = -2.197$$

where $\text{bracket}_{\text{meas}} = 0.2197$ is taken from direct measurement. All structural components are derived; the numerical value of β is reproduced to 0.7%.

3 What Remains: The Bracket Identity

Open: $\text{bracket} = \kappa / \ln 2$ as $N \rightarrow \infty$

The bracket is:

$$\text{bracket} = H_{\text{after}} \ln(1+M+\kappa) - H_{\text{before}} \ln(1+M)$$

Measured: 0.2197. Conjecture: $\kappa / \ln 2 = 0.2164$. Gap: 1.5%.

For this to equal $\kappa / \ln 2$, we need:

$$\frac{H_{\text{after}}}{1+M} + \frac{\delta H \cdot \ln(1+M)}{\kappa} = \frac{1}{\ln 2}$$

where $\delta H = H_{\text{after}} - H_{\text{before}}$.

At $N = 500$: LHS = 1.517, RHS = $1 / \ln 2 = 1.443$. Ratio: 1.051. The 5% discrepancy in this form reflects the 1.5% bracket gap amplified by the decomposition.

This identity requires knowing the joint distribution of $(H_{\text{before}}, H_{\text{after}}, M)$ in the $N \rightarrow \infty$ limit — specifically, how T_{avg} and H_{before} scale with N . This is a derivable calculation but is not completed today.

3.1 Why the Gap Is Likely Finite- N

The bracket gap of 1.5% at $N = 500$ is consistent with a correction of order:

$$\epsilon_N = \left(\frac{n_d}{N} \right)^{1/d_s} = \left(\frac{8}{500} \right)^{1/3} = 0.252$$

scaled by a coefficient of order 0.06, giving $0.252 \times 0.06 \approx 0.015$ — matching the observed gap. This is consistent with the interpretation that the bracket identity becomes exact as the domain boundary fraction $f_b \rightarrow 0$.

4 Full Status Table

Result	Status	Note
$K_{\text{eff}} < K_c$ (sub-threshold)	Derived	$0.0069 < 0.0637$
$\phi_{\text{var}} = 1/2$	Derived	Uniform phase in sub-threshold
$f_{12} = d_s/2 = 3/2$	Derived	1.3% at $N=500$
$H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P}_{\text{nb}} U)$	Derived	0.9%
$\rho = 1 - \frac{1}{2}(n_d/N)^{1/d_s}$	Derived	Domain boundary geometry
$\beta = f_{12} \cdot \text{bracket}/(-\kappa) = -2.197$	Derived	0.7% from measured
$\text{bracket} = 0.2197$	Confirmed	Direct measurement
$\text{bracket} \rightarrow \kappa/\ln 2 = 0.2164$	Watching	1.5% finite- N gap
CRF AI “bracket $\rightarrow 1/\ln 2$ ”	Corrected	Dimensional error; κ was missing
$ \beta = d_s/(2 \ln 2)$ (formal)	Open	Requires bracket proof

5 What Closes #21

The Remaining Step

Prove that in the $N \rightarrow \infty$ limit:

$$H_{\text{after}}(N) \ln(1+\bar{M}) - H_{\text{before}}(N) \ln(1+\bar{M}) \rightarrow \frac{\kappa}{\ln 2}$$

where $\bar{M}$ is the mean mass at the time of R-events.

This requires showing that the combination $\delta H(N) \cdot \ln(1+\bar{M}(N))$ converges to $\kappa/\ln 2$ as boundary-fraction corrections vanish.

Given the 1.5% gap and its apparent N^{-1/d_s} scaling, this is consistent with but not yet proven. A proof would make $\beta = -d_s/(2 \ln 2)$ a theorem of CRF with no free parameters.

The dimensional error is not a failure. It is the framework’s own integrity: κ does not cancel prematurely. It stays in the bracket, reminding us that the price of one resolution is not $1/\ln 2$ in the abstract, but $\kappa/\ln 2$ — one unit of material coupling, measured in bits. The algebra is not yet finished. Finishing it would make κ unnecessary as an independent parameter.